ANOOP SENTHIL

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Summary

AI/ML Engineer with an MSc in Artificial Intelligence and hands-on experience across LLM engineering, machine learning, computer vision, and AI systems. Experienced in designing, training, and evaluating intelligent systems, with expertise spanning LLMs, RAG, multimodal models, model fine-tuning, and scalable software development. Interested in building reliable AI systems that translate advanced ML research into practical, production-ready applications.

Education

Master of Science, Artificial Intelligence — **Sep 2025 - Sep 2026**
Queen Mary University of London — *London, United Kingdom*
- **Achievements:** Recipient of the Global Talent Scholarship (£5,000) for academic excellence
- **Coursework:** Machine Learning, Neural Networks, NLP, Information Retrieval, AI Ethics

Bachelor of Technology, Computer Science and Engineering — **Jul 2021 - Jul 2025**
Vellore Institute of Technology — *Vellore, India*
- **Coursework:** Data Structures & Algorithms, Database Management, Cloud Computing, Software Engineering

Projects

Hybrid RAG System for Scientific Question Answering | GitHub — **Aug 2026 - Sep 2026**
- Built a hybrid retrieval-augmented generation pipeline over the SciFact corpus, combining sparse BM25 first-stage retrieval with ColBERT-style late-interaction re-ranking to supply grounded, citation-backed answers from a 4-bit quantized Qwen2.5-3B large language model.
- Implemented a two-stage pipeline where BM25 surfaces top-100 candidates for ColBERT MaxSim re-ranking before the top-5 passages go to Qwen for cited generation, with the ColBERT projection layer fine-tuned via InfoNCE on BM25-mined hard negatives to sharpen domain-specific relevance ranking.
- Evaluated hybrid retrieval on 300 test queries, achieving 0.6858 nDCG@10, 0.6479 AP@10, and 0.6581 RR@10, with a re-ranking latency of 43.1ms/query.

Autonomous Equity Research & Valuation Agent | GitHub — **Apr 2026 - Sep 2026**
- Built a multi-agent financial research application with CrewAI and Groq, orchestrating natural-language ticker queries across Python-based market data, RSS news, SEC filings, and automated ReportLab PDF generation.
- Developed AI agents for document Q&A, bull/bear analysis, sentiment and risk assessment, alongside a 10-year FCF DCF valuation model using WACC and rule-based stock recommendations.
- Built a Qdrant-based RAG pipeline over SEC 10-K Item 1A disclosures using MiniLM embeddings and containerized the application with Docker Compose, with GPT-OSS 20B LLM inference served through Groq.

Evolutionary-Augmented Image-Text Retrieval System | GitHub — **Oct 2025 - Jan 2026**
- Designed a lightweight multimodal search system retrieving semantically relevant text labels for any input image, enabling accurate visual search without large pre-trained foundation models or expensive retraining.
- Constructed a custom Skip-Gram language backbone trained on a Visual Genome knowledge graph, expanded via a (1+λ) evolutionary algorithm to integrate CIFAR-100 labels while preserving the Skip-Gram's semantic structure.
- Aligned MobileNetV3 image features to the enriched text space using InfoNCE contrastive loss, achieving 91% Recall@10 on image-to-text retrieval with strong generalization across related categories.

Preprints

ReVA: A Region-Aware Visual Assistant for Visually Grounded Question Answering | arXiv | GitHub
- Developed ReVA to reduce object hallucination in visual question answering by explicitly grounding language generation in both global scene context and localized visual evidence, coupling CLIP ViT-L/14 Vision Transformer (ViT) with Qwen2.5-7B-Instruct LLM through separate whole-image and regional projection bridges.
- To provide the LLM with object-specific evidence for questions requiring fine-grained visual understanding, used RAM++ to identify question-agnostic objects present in the image and spaCy to extract question-dependent objects referenced by the question, Grounding DINO then localized these objects with bounding boxes, enabling RoI Align to extract their visual features from intermediate ViT representations.
- To integrate information across visual abstraction levels without discarding either low-level detail or high-level semantics, used cross-depth self-attention and multi-token pooling, followed by joint captioning and InfoNCE training and LoRA fine-tuning of the LLM to align visual representations with language.
- Evaluated ReVA on VQAv2, MMBench, POPE, and SEED-Bench benchmarks, achieving 73.79% VQAv2 accuracy and 82.85% mean F1 on POPE, while improving MMBench localization by 8.65 points and attribute comparison by 6.82 points.

Experience

DRDO, Ministry of Defence, Govt. of India — **Jan 2025 - May 2025**
AI Research and Development Intern — *Bangalore, India*
- Developed an image-to-code automation system using Computer Vision and Deep Learning to interpret control-law diagram images and automatically generate C code, eliminating manual coding for complex diagram inputs.
- Fine-tuned YOLO and OpenCV-based line detection to extract precise bounding boxes and line coordinates from noisy engineering diagrams, constructing accurate logical graphs for automated code generation within 10 seconds.
- Built a pipeline that processed images, translated them into graph representations, performed graph reasoning, and ran GCC syntax checks, which verified the correctness of the generated C code before deployment

Pivotrics — **Aug 2023 - Nov 2023**
Software Developer Intern — *Bangalore, India*
- Developed scalable RESTful APIs for a SaaS payment platform using Spring Boot, WebFlux, and PostgreSQL, with non-blocking async APIs and back-pressure management for data stream processing.
- Secured product and customer data by implementing JWT-based authentication and role-based access control using Spring Security.

Skills

- **Languages**: Python, Java, C, HTML/CSS, JavaScript, SQL
- **Frameworks & Libraries**: PyTorch, HuggingFace, LangChain, LangGraph, CrewAI, BMAD Method, OpenCV, spaCy, Spring Boot
- **Developer Tools**: Cursor, Claude Code, Git, GitHub, Jupyter, Docker
- **Databases & Cloud**: PostgreSQL, Qdrant, Amazon Web Services